2021 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Gravitation

A student is asked to calculate the gravitational field strength at a point above the surface of a planet.

(a)
Calculate the gravitational field strength at a height of 5.0 × 10^5 m above the planet's surface

Use the universal gravitation formula:

$$g = \frac{GM}{r^2}$$

Where:

- $G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
- $M = 4.8 \times 10^{24}\,\mathrm{kg}$
- $r = R + h = 6.4 \times 10^6 + 5.0 \times 10^5 = 6.9 \times 10^6\,\mathrm{m}$

$$g = \frac{6.67 \times 10^{-11} \times 4.8 \times 10^{24}}{(6.9 \times 10^6)^2} = \frac{3.2016 \times 10^{14}}{4.761 \times 10^{13}} = 6.72\,\mathrm{N\,kg^{-1}}$$

🧠 Revision Tips

- Always use $r = R + h$, where R is planet radius and h is height **above surface**
- Use the formula $g = \frac{GM}{r^2}$ only when far from the surface (i.e. non-uniform fields)
- Units of gravitational field strength are **N/kg** (same as m/s²)
- If you forget to add h to R, your answer will be too large — always check the context

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